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EU AI Act

Estonia

AI Ecosystem Brief

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AI & Partners

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AI & Partners defends and extends the digital rights of users at risk around the world. By combining direct technical support, comprehensive policy engagement, global advocacy, grassroots professional services, regulatory interventions, and participating in industry groups such as AI Commons, we fight for fundamental rights in the artificial intelligence age.

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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems are unknown

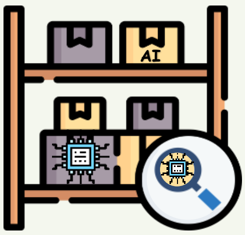
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

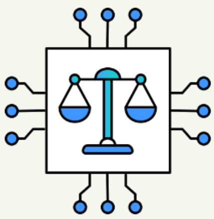
Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.

Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.

Contents

Overview

Artificial Intelligence (AI) is emerging as a transformative force across Estonia’s economy and society, with the country adopting a pragmatic and innovation-oriented approach to its governance. Rather than developing standalone national legislation, Estonia is aligning closely with the European Union’s AI Act, while supporting implementation through national strategies, public sector frameworks, and ethical guidelines to promote secure and trustworthy AI development.

At the heart of Estonia’s AI governance is its adherence to the EU AI Act. While Estonia initially considered national legislation, it decided to align its efforts with the broader EU regulatory framework for consistency. Therefore, Estonia does not have standalone national legislation exclusively governing AI, big data, or machine learning. Instead, these technologies are regulated under existing legal frameworks, including data protection laws, cybersecurity regulations, and various sector-specific statutes that are frequently updated. According to the Chief Information Officer of the Estonian government, Estonia aims to build upon the EU framework for responsible AI implementation.

Estonia's domestic AI ecosystem is supported by its national AI strategy, established in 2019 and regularly updated, which continues to drive the adoption of AI across both public and private sectors. The Ministry of Economic Affairs and Communications has an action plan for widespread AI adoption by 2030, with a planned investment of €85 million, as detailed in the White Paper on Data and Artificial Intelligence. Complementary strategic tools include initiatives like Bürokratt, an ambitious project to develop an integrated virtual assistant for secure and efficient access to government services.

The Estonian Information System Authority (RIA) published a comprehensive report on AI risks in July 2024, which also covers risk mitigation measures and offers practical guidance for enhancing cybersecurity in AI applications.

Institutional coordination of AI policy in Estonia remains in progress. While no single AI regulator has been designated, the White Paper on Data and Artificial Intelligence proposes the appointment of a state data architect, who will effectively serve as a Chief Data Officer for public-sector data resources. Public-sector bodies will also be required to create long-term strategies and action plans for data and AI use. Estonia’s data protection framework—anchored in GDPR compliance—is a critical element in governing AI-related use of personal data.

Internationally, Estonia is an active proponent of rules-based, ethical AI development. As an EU member state and a signatory to the OECD’s Principles on Artificial Intelligence (May 2019), Estonia advances a human-centric vision of AI that safeguards democratic values and digital rights. Estonia also engages in cross-border collaboration, such as its participation in the Finnish CSC coordinated LUMI pre-exascale supercomputer consortium, granting access to Europe's largest computing resource until 2025.

Estonia’s AI governance strategy also considers liability, civil responsibility, and data protection. In the absence of bespoke AI laws, liability is currently governed under existing statutes including the Estonian Copyright Act, the Personal Data Protection Act, and the GDPR. AI-generated works generally lack copyright protection unless a human author substantially influences the creative process. The Estonian judiciary has implemented AI tools to streamline legal processes, including automated solutions for anonymizing personal data in court decisions and automatic transcription of court hearings, highlighting the multi-dimensional implications of AI in the legal sector.

Through a blend of EU harmonization, strategic national programs, and evolving legal adaptation, Estonia is crafting an AI governance framework that supports technological innovation while embedding ethical and legal safeguards. Estonia's strategic AI plan states that while no major overhauls to the legal system are currently needed, some targeted legislative updates will be necessary for successful AI adoption.

Specific AI Governance or Law

Estonia has no standalone AI law but aligns with the EU AI Act and existing national laws on data protection and cybersecurity. Strategic documents guide responsible AI use, with ministries appointing AI leads and public-sector bodies required to draft implementation plans. No major national legislative changes are planned.

Specific AI Governance in Estonia

Estonia has been actively involved in shaping AI governance both domestically and at the EU level. While initially considering national legislation for AI systems, Estonia decided to align its efforts with the broader regulatory framework of the European Commission's EU AI Act to ensure consistency across Member States. Therefore, Estonia does not have standalone national legislation exclusively governing AI, big data, or machine learning. Instead, these technologies are regulated under existing legal frameworks, including data protection laws, cybersecurity regulations, and various sector-specific statutes that are frequently updated. According to the Chief Information Officer of the Estonian government, Estonia aims to build upon the EU framework for responsible AI implementation rather than creating its own. Despite this, Estonia has developed strategic documents to guide the development and deployment of AI, strongly encouraging AI innovation while safeguarding public interests. No substantial national legislative changes are currently planned for the future. An interesting development is that Estonian ministries are being tasked with appointing a dedicated person responsible for the sector by next year, and all public-sector organizations will be required to draw up an AI implementation plan. The Estonian government has taken measures to prevent adverse outcomes from AI.

AI Ethics Guidelines and Sectoral Governance Tools

Estonia aligns its regulatory framework with the EU AI Act, focusing on ethical AI deployment. In May 2019, the Estonian government signed the Organization for Economic Co-operation and Development's (OECD's) Principles on Artificial Intelligence, reflecting the principles of human-centric and ethical AI development recommended by the OECD. The Estonian Copyright Act, aligned with the EU Directive on Copyright in the Digital Single Market, governs the use of copyrighted works, databases, and AI-generated outputs. Databases are protected under a sui generis right if a substantial investment is made in obtaining, verifying, or presenting the data. Estonia's Personal Data Protection Act, implementing the General Data Protection Regulation (GDPR), ensures strict data privacy, and the Cybersecurity Act safeguards critical information systems. Contractual agreements are crucial for regulating data use, access, and exploitation. AI-generated works generally lack copyright protection in Estonia unless a human author substantially influences the creative process, meeting the originality threshold. If the protection criteria are met, copyright protection is granted automatically upon the creation of the original work, even if created with AI assistance.

Proposed AI Regulatory Authority and Future Outlook

At the time of writing, Estonia does not have a single designated regulator exclusively for AI, big data, or machine learning, as these technologies are regulated under existing legal frameworks. Instead, the government encourages AI innovation while safeguarding public interests. The Ministry of Economic Affairs and Communications has an action plan for widespread AI adoption across public and private sectors by 2030, with a planned investment of €85 million. This initiative is detailed in the White Paper on Data and Artificial Intelligence, which sets out strategic objectives for 2024–2030. The White Paper envisions Estonia becoming a global leader in data-driven governance and innovation by utilizing data more effectively and responsibly. Key proposals include a mature data economy by 2030, allowing seamless data selling, sharing, and reuse by both public and private organizations. Public-sector bodies will be required to create long-term strategies and action plans for data and AI use, supported by dedicated funding. A major target is to reduce citizens' administrative workload by 70% through improved data reuse. To oversee this, a state data architect will be appointed, acting as a Chief Data Officer for public-sector data resources. The White Paper also outlines changes for government interaction with businesses, aiming for single data submissions and automated interactions by 2030.

Discussion: Estonia's National AI Strategy

Estonia aligns its AI governance with the EU AI Act, avoiding standalone legislation. Its national strategy promotes ethical, human-centric AI, public-sector innovation, and digital inclusion. Key initiatives include public-service pilots, X-Road infrastructure, AI education programs, and risk mitigation efforts, positioning Estonia as a digitally advanced, EU-aligned AI leader.

Advancing AI in the public sector in estonia

Advancing AI in the private sector in estonia

Developing AI research and education in estonia

Shaping the legal environment for AI adoption in estonia

Analysis

Estonia's AI strategy is defined by its commitment to embedding artificial intelligence across both its public and private sectors. The country's e-government infrastructure, underpinned by the X-Road data exchange platform, is pivotal in this regard. The strategy aligns with the EU's regulatory trajectory, as Estonia decided to align its AI governance efforts with the EU AI Act to ensure consistency across Member States, rather than creating bespoke national legislation. This strategy also promotes national priorities such as data-driven governance, leveraging AI to automate public services, and digital inclusion. Central to Estonia's vision is its focus on human-centric AI and ethical deployment, actively promoted through its digital governance strategies. While not relying solely on regulation, Estonia leverages adaptive policy tools and a steering group, led by the Ministry of Justice and Digital Affairs (JDM) and consisting of government agencies and key stakeholders, to coordinate and monitor the implementation of its strategy.

A key feature of Estonia's governance model is its strategic alignment with EU values and global standards. Estonia's national AI strategy is explicitly synchronized with and supports relevant EU-level activities. In May 2019, Estonia also signed the OECD's Principles on Artificial Intelligence, reflecting its commitment to human-centric and ethical AI development. Estonia's approach acknowledges the need for value-based prioritisation in areas such as digital infrastructure and aims to become a global leader in data-driven governance and innovation, particularly through more effective and responsible data utilization.

Estonia supports AI experimentation and agile governance through measures such as establishing sandbox environments to experiment with AI solutions in the public sector. The government also facilitates the development and preparation of AI projects by agencies through brainstorming sessions, project development advice, matchmaking, and advice on funding opportunities. Although Estonia does not currently have formal AI sandboxes, the country's AI governance encourages pilot initiatives. For instance, a consent management pilot project was planned for June 2020. Emerging institutions and collaborative efforts are envisioned to attract global talent and reinforce national excellence. Estonia is also actively involved in EuroHPC projects to make high-performance computing capacity available to R&D institutions and companies.

AI governance in Estonia is deeply tied to digital competitiveness and public sector efficiency. The strategy targets the deep integration of AI across priority areas, including streamlining bureaucracy and improving user experience for citizens. This includes initiatives like Bürokratt, an integrated virtual assistant for government services. Estonia champions public-private partnerships and open data infrastructures. The development of data governance and increasing the availability of open data are ongoing activities. Estonia's e-government infrastructure, underpinned by the X-Road data exchange platform, is pivotal for secure and efficient data exchange, which powers ethical AI innovation. A strong emphasis on digital inclusion and lifelong learning runs through Estonia's AI strategy. The country prioritizes developing a multidisciplinary AI workforce through investments in education and training. For instance, in March 2025, Estonia launched the AI Leap initiative to integrate AI education into high schools, providing 20,000 students aged 16–17 with free access to AI learning tools and AI training workshops for 3,000 teachers. The objective is to strengthen students' critical thinking abilities and equip them for the changing employment landscape shaped by AI. This initiative will serve as the foundation for enhancing both Estonia's international economic competitiveness and the next stage of e-Estonia's development. Estonia also supports SME and startup development, with numerous funding opportunities and access to dedicated AI accelerator programs available for AI-driven businesses.

Estonia champions dynamic, future-oriented AI governance that balances innovation with resilience. This includes promoting trustworthy AI applications while guarding against risks. The Estonian Information System Authority (RIA) published a comprehensive report on AI risks in July 2024, which also covers risk mitigation measures and offers practical guidance for enhancing cybersecurity in AI applications. As the AI strategy progresses, Estonia continues to refine its implementation through stakeholder engagement, iterative policy design, and international cooperation. This strategic model—combining EU harmonisation, domestic capacity-building, and global ethical alignment—positions Estonia as a leader in shaping a trustworthy, inclusive, and future-ready AI ecosystem.

Estonia's Approach to AI Regulation and Governance

Estonia has no standalone AI law but follows the EU AI Act as its primary framework. Its national strategy, updated since 2019, promotes ethical AI and innovation. Ministries must appoint AI leads, and public bodies must draft implementation plans. €85 million is earmarked for AI adoption across sectors by 2030.

Regulatory and Strategic Frameworks

Estonia's AI governance is closely integrated within the European Union's regulatory framework. Rather than enacting bespoke national legislation for AI, Estonia has opted to align its efforts with the EU AI Act, which will serve as the primary legal framework for AI systems across the EU. This approach ensures consistency across Member States.

While standalone national AI legislation is not being pursued, Estonia has implemented strategic documents to guide AI development and deployment, promoting innovation while safeguarding public interests. According to the Chief Information Officer of the Estonian government, the country aims to build upon the EU framework for responsible AI implementation. All public-sector organizations will be required to develop AI implementation plans, and ministries are tasked with appointing dedicated AI representatives by next year. Existing legal frameworks, including data protection laws, cybersecurity regulations, and various sector-specific statutes, are regularly updated to encompass AI.

Estonia's AI strategy, established in 2019 and regularly updated, remains the cornerstone of its AI vision. The Ministry of Economic Affairs and Communications has an action plan for widespread AI adoption across public and private sectors by 2030, with a planned investment of €85 million, as detailed in the White Paper on Data and Artificial Intelligence.

Institutional Support and Strategic Missions

Estonia's AI governance is coordinated through ministerial collaboration, with key roles played by ministries involved in economic affairs and communications. While Estonia does not have a single designated AI regulator, the government is focusing on creating a mature data economy by 2030 and has proposed the appointment of a state data architect, who will effectively serve as a Chief Data Officer for public-sector data resources. This indicates a growing institutional focus on comprehensive data and AI governance.

Strategically, AI governance is embedded in Estonia's broader digital transformation initiatives, such as Bürokratt, an ambitious project to develop an integrated virtual assistant for secure and efficient access to government services. These frameworks emphasize the secure and efficient adoption of AI, with a focus on streamlining bureaucracy, enhancing user experience, and promoting a secure digital environment.

Standards Development and Global Engagement

Estonia actively contributes to the development of EU and international AI standards. In May 2019, the Estonian government signed the Organization for Economic Co-operation and Development's (OECD's) Principles on Artificial Intelligence, reflecting its commitment to human-centric and ethical AI development.

The Estonian Information System Authority (RIA) published a comprehensive report on AI risks in July 2024, providing practical guidance for enhancing cybersecurity in AI applications and offering risk mitigation measures. Although Estonia has not yet issued national AI ethics legislation, public authorities and research institutions actively shape ethical guidance, particularly on data governance and algorithmic fairness.

Sectoral and Extended Regulatory Approaches

AI deployment in Estonia is increasingly subject to sectoral regulation and horizontal legal frameworks. For instance, the Estonian judiciary has implemented AI tools for automated anonymization of personal data in court decisions and automatic transcription of court hearings. These developments align with the country's commitment to digitalizing legal processes.

Existing statutes, including the Estonian Copyright Act, the Personal Data Protection Act, and the Cybersecurity Act, provide important guardrails for algorithmic decision-making, surveillance, and data processing. While AI-generated works generally lack copyright protection in Estonia, copyright can be granted if a human author substantially influences the creative process. In the private sector, GDPR compliance remains a foundational legal obligation for AI systems involving personal data.

Fostering Innovation and Indigenous Development

Estonia views AI as a strategic technology for national competitiveness and sustainability. The government is investing in research, public-private partnerships, and pilot projects to scale AI innovation across sectors, aiming for 75% of private enterprises to have integrated AI technologies to boost productivity by 2030. Estonia's thriving start-up ecosystem, supported by strong R&D partnerships and a progressive regulatory environment, is a key driver of AI innovation. The government also promotes initiatives such as launching a public online course to increase general AI literacy and establishing sandbox environments to experiment with AI solutions in the public sector.

The private sector benefits from access to innovation and development grants specifically intended for projects involving machine learning technologies. The Estonian government has also launched a €100 million fund to support the defence tech sector, aiming for the industry to generate €2 billion in revenue by 2030.

AI in Public Sector Transformation

AI is playing an increasing role in transforming Estonia's public administration. Key initiatives include Bürokratt, designed to provide secure and efficient access to government services through an integrated virtual assistant. The White Paper on Data and Artificial Intelligence aims to reduce citizens' administrative workload by 70% through improved data reuse by 2030, indicating a strong commitment to AI-driven public service automation. Public-sector bodies will be required to create long-term strategies and action plans for data and AI use.

AI Skills, Education and Inclusion

Estonia emphasizes human capital development in its AI strategy, with a strong focus on education, upskilling, and digital literacy. Integrating data literacy and management into all levels of education is a key objective of the White Paper on Data and Artificial Intelligence. This ensures that workers and citizens can actively participate in an AI-enabled society. The country also supports initiatives such as a public online course to increase general AI literacy.

Infrastructure, Compute and Data Foundations

To enable effective AI deployment, Estonia is investing in digital infrastructure, secure data access frameworks, and leveraging its established X-Road data exchange platform. The White Paper on Data and Artificial Intelligence proposes a mature data economy by 2030, allowing seamless data selling, sharing, and reuse by both public and private organizations. While specific AI liability insurance details are not extensively covered, existing legal frameworks for redress are in place. The EU AI Act will also require Estonia to establish mechanisms for real-world testing under regulatory oversight, potentially leading to formalised AI sandboxes.

Estonia's Approach to Fostering AI Innovation

Estonia's AI strategy targets broad adoption of AI and big data across sectors by 2030, guided by the Ministry of Economic Affairs and Communications. Backed by €85 million and outlined in the White Paper on Data and Artificial Intelligence, the plan envisions Estonia as a global leader in data-driven innovation.

Unlocking Economic Value through Strategic AI Deployment

Estonia's AI strategy drives the adoption of AI, machine learning, and big data across both public and private sectors. The Ministry of Economic Affairs and Communications has an action plan for widespread AI adoption by 2030, supported by a government investment of €85 million. This initiative, detailed in the White Paper on Data and Artificial Intelligence, aims for Estonia to become a global leader in data-driven governance and innovation. By 2030, Estonia's data economy is projected to be mature enough for seamless data selling, sharing, and reuse by firms.

Driving Innovation via Research Clusters and Application Hubs

Estonia fosters AI innovation through a thriving start-up landscape and strong research and development partnerships. The government actively supports pilot projects and aims to accelerate AI adoption across public institutions and the broader economic landscape. This includes initiatives such as establishing sandbox environments to experiment with AI solutions in the public sector and broader initiatives such as testbed for companies to safely test new technologies and business models which are prohibited by current regulations. Additionally, the private sector benefits from innovation and development grants specifically intended for projects involving machine learning technologies. The private sector also benefits from AI development programmes, which are designed to help Estonian businesses adopt AI solutions, making them more competitive and ensuring the sustainable success of the Estonian economy in the future.

Enabling Trusted and Responsible Data Use

Estonia's e-government infrastructure, particularly the X-Road data exchange platform, is pivotal for secure and efficient data exchange. The White Paper on Data and Artificial Intelligence emphasizes secure, human-centered AI systems and aims for transparent and effective data utilization. Public-sector bodies will be required to create long-term strategies and action plans for data and AI use, supported by dedicated funding. A state data architect will be appointed to manage public-sector data resources.

Scaling AI in Public Administration

AI plays a significant role in automating public services to streamline bureaucracy and improve user experience for citizens. A key initiative is Bürokratt, an integrated virtual assistant for secure and efficient access to government services. The government aims to reduce citizens' administrative workload by 70% through improved data reuse. The Estonian judiciary has also embraced digital transformation by implementing AI tools to streamline legal processes, including automated solutions for anonymizing personal data in court decisions and automatic transcription of court hearings. The government aims to fully digitize the court system by the end of 2025.

Advancing Compute and Research Infrastructure

Estonia actively participates in projects like the Finnish CSC coordinated LUMI pre-exascale supercomputer consortium, granting access to Europe's largest computing resource until 2025. This initiative is part of Estonia's broader strategy to make high-performance computing capacity available to R&D institutions and companies. Estonia's strategic AI plan also includes fostering research and development (R&D) base.

Empowering SMEs and Startups

Estonia's thriving start-up landscape and forward-thinking regulatory environment help drive AI innovation. Numerous funding opportunities, including incubator programs, are available to support AI-driven businesses. The government has also launched a €100 million fund to support the defense tech sector, aiming for the industry to generate €2 billion in revenue by 2030. By 2030, 75% of private enterprises are expected to have integrated AI technologies to boost productivity.

Developing a Future-Ready Workforce

Estonia emphasizes increasing relevant skills and R&D base as part of its AI strategy. In March 2025, Estonia launched the AI Leap initiative to integrate AI education into high schools, providing 20,000 students aged 16–17 with free access to AI learning tools and offering AI training workshops for 3,000 teachers. The primary objective is to strengthen students' critical thinking abilities and equip them for the changing employment landscape shaped by AI. The strategy includes integrating data literacy and data management topics into all levels of education.

Aligning AI with Sustainability Goals

The White Paper on Data and Artificial Intelligence includes the development of centralized, data-driven tools, such as sustainability and viability assessment platforms, using state-collected data. This aligns AI with broader sustainability objectives.

Building Public Trust and Ethical AI

The Estonian government actively promotes the ethical and responsible deployment of AI technologies through its digital governance strategies. In May 2019, the Estonian government signed the OECD's Principles on Artificial Intelligence, reflecting its commitment to human-centric and ethical AI development. Efforts are underway to increase public awareness of AI through online courses.

Shaping Global AI Governance and Standards

Estonia has been actively involved in shaping AI governance both domestically and at the EU level. While initially considering national AI legislation, Estonia decided to align with the EU AI Act to ensure consistency across Member States. Estonia's national AI strategy is synchronized with and supports relevant EU-level activities.

> *'Estonia's AI strategy promotes ethical innovation, boosts public-private AI adoption, supports startups and research, and prepares its workforce—positioning the country as a digital leader aligned with the EU AI Act.'*

Figure 1: Estonia Action Plan

Knowledge sharing & awareness

- Events to introduce AI to agencies
- Senior management training
- Thematic network on data science & AI
- Dissemination of success stories
- Public awareness online course
- Presentations and experience exchange

Funding and Innovation Support

- Existing innovation and product development grants
- AI-specific pilot project support (TRL 4–6 and 5–7)
- New €50M digitalisation measure (2019–2021)
- Expansion to other economic sectors
- Innovation competition for AI on government data

Talent and Capacity Development

- Hiring lecturers to teach AI cross-disciplinarily
- Targeted AI specialisation for MSc and PhD students
- Continued applied research funding (IT Academy)
- Creating organisational capacity for industry–research collaboration

Developing the legal environment for AI

- Draft and submit "Kratt Law" legal package
- Targeted legal changes, no fundamental legal overhaul needed

TI-HÜPE

AI Leap 2025: Together with this, Estonia announced *AI Leap 2025* (TI-Hüpe 2025), a national initiative to integrate AI into its education system, launching on 1 September 2025. Building on the legacy of the Tiger Leap program that digitized Estonian schools in the 1990s, AI Leap aims to make Estonia the world's smartest nation by equipping students and teachers with access to leading AI tools and the skills to use them effectively. The initiative will begin with 20,000 high school students and 3,000 teachers, expanding in 2026 to include vocational schools and more students. Backed by the government, private sector leaders, and global AI companies like OpenAI and Anthropic, AI Leap will be coordinated by a dedicated foundation and include comprehensive teacher training and international cooperation.

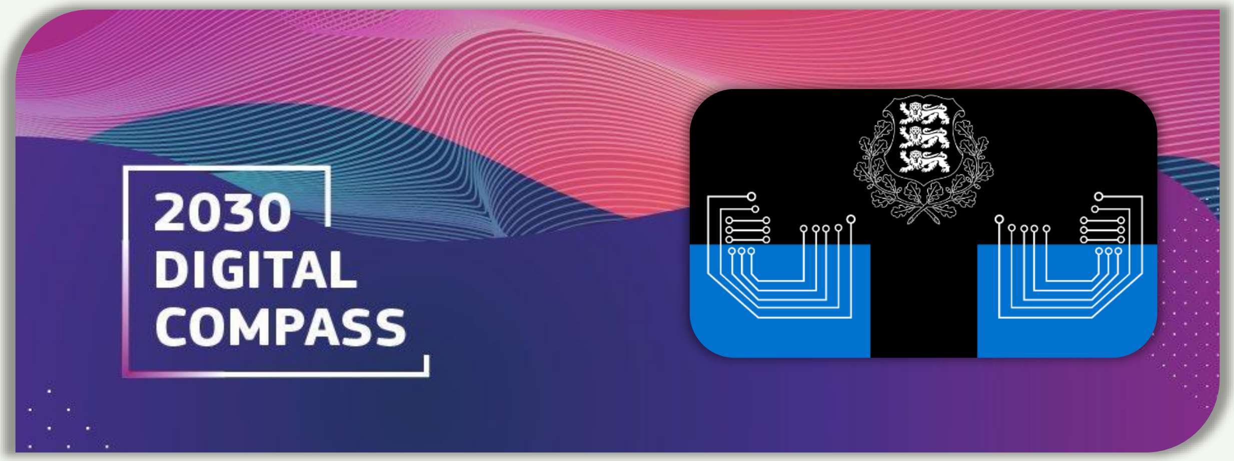
Digital Decade 2025: Country Report

'In its adjusted roadmap, Estonia proposed a new target and trajectory for the adoption of AI in line with the EU target of 75%.'

As mentioned in Estonia's adjusted roadmap, the Technopol AI Development Programme, launched in 2022–2025, continues to guide the AI scene. The programme focuses on industrial companies and start-ups in the environmental, health, and energy sectors. The AI start-up incubator has launched 30 enterprises, and more than 19 new AI-based products/enterprises have entered the market. The programme will continue until 2027, with plans to support another 30 start-ups.

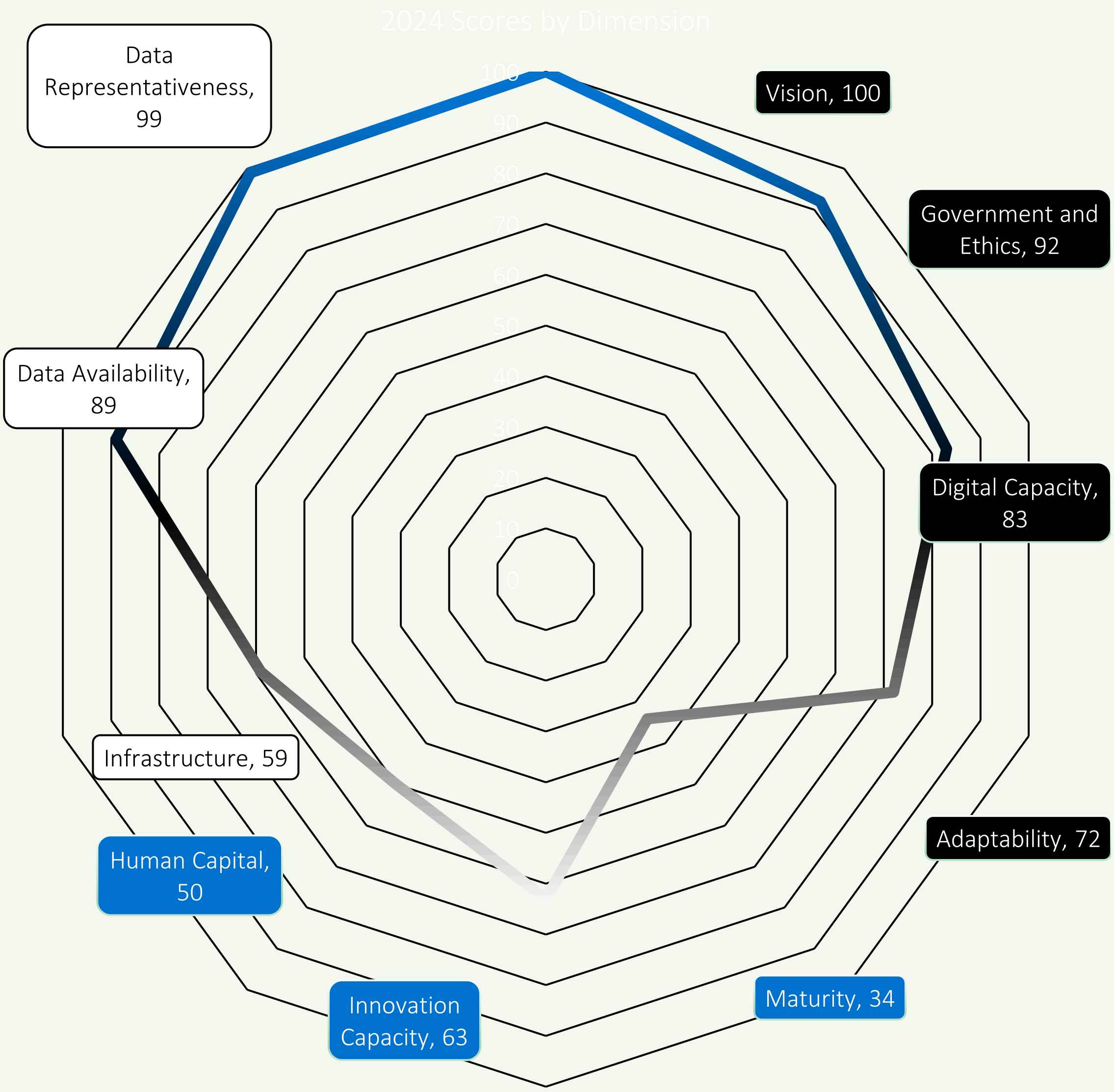
Estonia is also promoting the use of AI in the public sector. AI adoption support was provided to over 30 public sector organisations, and 172 AI-based solutions have been implemented. The adoption of generative AI has been prioritised through more than 15 pilot projects.

Estonia is placing an emphasis on preserving its language by integrating Estonian into AI models. AI language models face difficulties in learning small languages due to limited data. To address this, Estonia has shared nearly four billion linguistic datasets with a leading tech company in the digital sphere to improve functions such as voice assistants, chatbots, and translation tools.



Government AI Readiness by Oxford Insights

'Eastern Europe ranks 4th in this year's Government AI Readiness Index, with an average score of 57.88—over 10 points above the global average (47.59). Estonia (72.62) leads the region and is the only Eastern European country to feature in the global top 25. The region performs above the global average across all three pillars. Its strongest performance is in the Data & Infrastructure pillar (72.10), making it the third-best performing region globally in this area. Eastern Europe also shows solid results in the Government pillar (62.52), where it again ranks third worldwide. However, the Technology Sector remains a challenge, with a regional score of 39.03—more than 20 points behind the other two pillars—and ranking fifth globally, falling narrowly behind the Middle East and North Africa (39.34). This reflects the need for further investment in technological capacity and innovation to unlock the region's full AI readiness potential. Eastern Europe's strong governance frameworks and robust data infrastructure continue to drive its progress, positioning it as a leader among emerging regions in AI readiness.'



Government

Estonia scores 86.71 in the Government pillar, reflecting an exceptionally strong performance that significantly contributes to its leading position within Eastern Europe and its global top 25 ranking. This indicates robust governance frameworks and a high level of preparedness for AI implementation. Estonia's proactive stance in preparing for the EU AI Act, appointing dedicated working groups for implementation, and having existing regulations concerning automated decision-making in the public sector underscore this strong performance. Its emphasis on digital transformation and ethical AI usage within government operations further solidifies its standing.

Technology Sector

Estonia scores 48.97 in the Technology pillar. While this is the lowest-scoring pillar for Estonia, it still outperforms the Eastern European regional average (39.03) by nearly 10 points. This suggests a relatively stronger, though still developing, technological capacity compared to the rest of the region. Despite being behind the other two pillars, Estonia's vibrant startup scene, coupled with government support for R&D and innovation, indicates efforts to address this area. Further investment in scaling technological capacity, fostering innovation, and accelerating venture capital will be crucial for Estonia to unlock its full AI readiness potential in this sector.

Data & Infrastructure

Estonia achieves an impressive 82.19 in the Data & Infrastructure pillar, making it a top performer not only within Eastern Europe but also globally. This strong performance is driven by its robust digital infrastructure, notably the X-Road data exchange platform, which facilitates secure and efficient data sharing. Estonia's commitment to data protection (aligning with GDPR), initiatives for ethical and interoperable data infrastructures, and active participation in data-driven value creation highlight its foundational strength in supporting and leveraging data for AI development and deployment. This positions Estonia as a leader in its ability to support AI.

Estonia leads Eastern Europe with a total AI Readiness Index score of 72.62, positioning it as the only country from the region in the global top 25. Its remarkable performance is driven by its exceptionally strong governance frameworks and robust data and infrastructure, which significantly outperform the regional averages. While the Technology Sector, though relatively strong for the region, remains an area for continued investment, Estonia's overall progress underscores its leadership in AI readiness among emerging economies. Its strategic focus on digital transformation, ethical AI, and alignment with EU objectives continues to drive its advancement.

Conclusion

Estonia's artificial intelligence ecosystem is entering a decisive phase—building on strong governance, digital infrastructure, and strategic foresight to realize a human-centric AI future. Anchored by its national AI strategy (initiated in 2019) and aligned with the EU AI Act, Estonia's approach reflects its national values of openness, trust, and technological sovereignty. By embedding AI into broader digital policy goals, Estonia aims to solidify its position as a global leader in digital transformation.

Key programs such as Bürokratt (an integrated virtual assistant for government services) , its well-established e-government infrastructure underpinned by the X-Road data exchange platform , and active participation in EuroHPC projects to access high-performance computing collectively establish Estonia's strategic groundwork. These initiatives position AI not only as a technological tool but as a lever for public sector transformation and social equity.

Yet, pressing challenges remain. Despite high scores in government readiness (86.71) and data infrastructure (82.19) , Estonia's relatively modest technology score (48.97) underscores the need to accelerate the scale-up of commercial AI applications. Barriers include the need for further investment in technological capacity and innovation. Bridging these gaps will require more robust support mechanisms—such as further developing regulatory sandboxes and AI experimentation environments. Estonia already has a steering group to coordinate and monitor the implementation of its AI strategy.

Estonia's legal and regulatory framework is evolving in step with the EU AI Act. While Estonia does not have standalone national AI legislation, it has decided to align its efforts with the EU AI Act to ensure consistency across Member States. Ministries are being tasked with appointing a dedicated person responsible for the sector by next year, and all public-sector organizations will be required to draw up an AI implementation plan. In the interim, existing legal frameworks, including data protection laws (like GDPR) and cybersecurity regulations, govern AI technologies. The Estonian Information System Authority (RIA) published a comprehensive report on AI risks in July 2024, offering practical guidance for enhancing cybersecurity in AI applications.

To maintain public trust and social legitimacy, Estonia emphasizes transparent and ethical AI deployment. The government actively promotes the ethical and responsible deployment of AI technologies through its digital governance strategies. Estonia is a signatory to the OECD's Principles on Artificial Intelligence. Measures to prevent adverse outcomes from AI have been taken, and Estonian courts are using automated AI solutions to remove personal data from decisions to ensure efficiency.

Estonia's commitment to inclusive innovation is another strategic strength. Through initiatives like the AI Leap program, which integrates AI education into high schools , the country is investing in a diverse AI-capable workforce. The government aims to strengthen students' critical thinking abilities and equip them for the changing employment landscape shaped by AI. Support for SMEs and startups is provided through various funding opportunities and incubator programs. Internationally, Estonia plays a catalytic role in shaping AI norms and interoperability standards. As an EU member state, Estonia's national AI strategy is synchronized with and supports relevant EU-level activities. Estonia is well-positioned to contribute to global discourse on AI policy and law.

In sum, Estonia's AI strategy is coherent, values-driven, and institutionally well-grounded. The foundations are robust: cross-sectoral collaboration, forward-looking infrastructure, and EU-aligned governance. With continued focus on scaling up its technology sector and further developing commercial AI applications, Estonia can move from being a thoughtful AI adopter to a bold global standard-setter in human-aligned AI systems.

Appendix A: Expert Vignettes

Emanuele Picariello

Offensive Security Engineer

"Every time I spot those Bolt-Starship rovers rolling through Tallinn, reminds me that the city is a living AI playground, surrounded by unicorns like Bolt, Veriff, and Glia. I wouldn't be surprised if AI automation catches on across Europe sooner than expected."

Biography

Emanuele Picariello, From a young age, I've been attracted by the dual nature of technology: its immense potential to transform lives and its vulnerabilities that demand constant vigilance. This curiosity led me to specialize in cybersecurity, earning certifications like OSCP, OSEP, CRTO, and more. Along the way, I've collaborated with businesses to safeguard their systems, building resilience in an evolving digital landscape.

Pablo Gómez

ePlaneAI

"Estonia shows that in the age of AI, success isn't measured by size, but by strategy, digital readiness, and smart execution. With its advanced digital infrastructure, global mindset, and ability to move fast and collaborate closely, Estonia is turning its compact scale into a competitive advantage, proving that even a small nation can lead the world in AI innovation."

Biography

Pablo Gómez, is a data wizard and his job is to help businesses, like yours, make sense of the data they create. Pablo's experience covers businesses of all sizes - from startups to big multinational corporations, and across different fields like Marketing, Finance, E-commerce, and Agribusiness. He's served as a trusted advisor to CEOs, helped C-level executives with data strategy, and have also been a team player in larger groups.

Karin Jaanson

Estonian Research Council

"Estonia's digital-first mindset and world-class tech talent make it a natural launchpad for AI innovation. Also with a sucess of e-government and a culture that embraces experimentation, Estonia is so ready for AI to shape its future."

Biography

Karin Jaanson is Executive Director of the Estonian Research Council, leading strategic and administrative efforts since 2015. She has extensive experience in public administration, research policy, and economic development, with prior roles in government and academia. She holds degrees in economics and public administration from universities in Estonia and the USA.

Dr. Ott Velsberg

Government Chief Data Officer

> *"A truly competitive and sovereign Europe will be built on data and AI systems that respect human agency, enable trust, and work across borders—turning our values into a driver for innovation."*

Biography

Dr. Ott Velsberg is a renowned Estonian IT specialist and government official, known for his expertise in data governance and data science. As the current Chief Data Officer of Estonia, he is responsible for driving the country's data and AI policy and initiatives related to the use of data in the public sector. Under his leadership, Estonia has gained a reputation as a trendsetter in data governance, open data, and artificial intelligence.

Doris Põld

ITL Estonia

> *"Estonia scores 48.97 in the Technology pillar. While this is the lowest-scoring pillar for Estonia, it still outperforms the Eastern European regional average (39.03) by nearly 10 points. This suggests a relatively stronger, though still developing, technological capacity compared to the rest of the region. Despite being behind the other two pillars, Estonia's vibrant startup scene, coupled with government support for R&D and innovation, indicates efforts to address this area. Further investment in scaling technological capacity, fostering innovation, and accelerating venture capital will be crucial for Estonia to unlock its full AI readiness potential in this sector."*

Biography

Doris Põld, Experienced CEO with a demonstrated history of working in the ICT industry. Promoting the cooperation of Estonian IT and telecommunications companies and organizations in Estonia's development towards information society. Skilled in team management, coaching cluster/project development and team building. Building collaborative networks between competitors, setting up their common goals and working towards that is my strongest point.

Priit Lätt

TEGOS Legal

*The Estonian Bar Association's **AI guidance** issued on 3 December 2024 exemplifies a proactive and responsible approach to AI in the legal profession, setting a high standard among European legal institutions. Only a few bar associations in Europe have adopted comparable guidelines, underscoring Estonia's leadership in addressing the ethical and practical challenges posed by AI. Crucially, the Estonian guidelines affirm that responsibility for legal advice remains with the lawyer, not shifted to any AI tool. This forward-looking stance by the Estonian Bar highlights Estonia's commitment to **innovative yet principled** AI use in legal practice, serving as a model for others in Europe.*

Biography

Priit Lätt is a visionary at the intersection of law and technology, guiding both public institutions and tech companies through uncharted legal terrain. He has led landmark cases, including Estonia's largest software litigation and the country's first Bitcoin-related lawsuit, shaping legal thinking in the digital era. Beyond litigation, Priit helped draft Estonia's first standard contract for public-sector software procurement, creating a blueprint for AI and tech agreements nationwide. A thought leader in AI and IT law, he shares his insights through lectures, publications, and industry forums, bridging innovation and regulation while shaping Estonia's digital future.

Andres Lehtmets

InsurTech4Good.com

“Drawing on my experience shaping digital finance policy and regulation at European and global level, I see Estonia as uniquely positioned to lead. With its e-state infrastructure, agile regulation, and commitment to trust and human-centricity, Estonia offers a uniquely enabling environment to test, start, and scale AI-powered financial and insurance solutions. Notably, financial and insurance sectors are already the most active users of AI technologies in Estonia – a clear signal that innovation is both real and responsible here”

Biography

Andres Lehtmets is the founder of InsurTech4Good.com, a consultancy providing regulatory strategy, thought leadership, research, training, and policy advice to financial institutions, FinTechs, and regulators worldwide. His focus areas include AI regulation (notably the EU AI Act), open finance and open insurance, new business models, and EU digital finance policy. He previously worked at the European Insurance and Occupational Pensions Authority (EIOPA), where he led work on digitalization, and earlier in the Insurance Policy Department at the Estonian Ministry of Finance. He has published academic papers on financial innovation in peer-reviewed journals and is a frequent speaker at industry events and executive training programmes.

Sirli Heinsoo

Majandus- ja Kommunikatsiooniministeerium

> *Structured business data is the backbone not only for entrepreneurship, but also for the future public services. When reused intelligently across institutions and combined with AI, it enables timely and tailored support for businesses.*

Biography

Sirli Heinsoo is dedicated to unlocking the value of high-quality business data to reduce administrative burden for entrepreneurs. As Director of the Digital Economy Department, she leads efforts to improve business data quality and support digitalisation within and between companies, coordinate the shift to data-driven reporting across institutions, and develop public services for businesses that are more user-friendly, proactive, and personalised..

Appendix B: Spotlight – Estonian AI Research

Estonia currently hosts **10 Centres of Excellence**, which are collaborative research consortia composed of internationally recognized research groups. Their goal is to enhance the quality, efficiency, and impact of scientific research through interdisciplinary cooperation. These centres focus on areas critical to Estonia's development and global competitiveness.

The Centres of Excellence support Estonian top-level research to strengthen the position of Estonian research cooperation and competitiveness in Europe. Currently there are 10 Centres of Excellence.

Estonian Centre of Excellence in AI

The Centre of Excellence develops innovative methods to support the creation of trustworthy artificial intelligence systems. The methodology promotes AI capabilities in areas important to Estonia, such as e-governance, healthcare, business process management, and cybersecurity.

"The use of foundation models, such as large language models, enhances AI's ability to interpret complex data and generate solutions. The Centre aims to advance novel methodologies for building effective and reliable analysis and prediction systems using foundation models; implement control mechanisms and constraints to ensure complex AI systems comply with specifications; adapt and improve AI systems to enhance application efficiency; and ensure the security and privacy of AI systems."

zProf Meelis Kull – Head of EXAI University of Tartu

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